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Camera-Only,Continuous Real-Time Marathi Sign Language Recognition: A Systematic Literature Review of Hybrid 1D-CNN–BiLSTM–Temporal Attention Architectures

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Abstract

Background: Sign language is used as a mode of communication among the majority of the deaf and hard of hearing individuals in Maharashtra state of India. But there is hardly any research on automatic recognition of Marathi Sign Language (MSL).Most of the existing systems for continuous sign language recognition (CSLR) employ vision-based techniques that recognize isolated signs from static images under controlled environments. Such systems do not work effectively in real-world scenarios like hospitals, classrooms and government offices where sign language communication takes place continuously. Methods: This paper presents a systematic literature review (SLR) of continuous sign language recognition (CSLR) systems with respect to the architecture, feature modality, datasets used and real-time deployment. The study adheres to PRISMA guidelines. Out of 140 papers initially retrieved on July 28– Aug 5, 2026 through keyword searches on 8 digital libraries, a total of 15 relevant studies with average quality score of 9. 0/10 are reviewed after applying inclusion/exclusion criteria & quality appraisal. Findings: Majority of research work on CSLR employs appearance-based convolutional neural networks (CNN) pipeline for recognition. Though, hybrid CNN—Long Short Term Memory (LSTM) and Transformer based sequence models are increasingly being used for sequence-level recognition, these models suffer from high computational cost. None of the reviewed articles addresses the problem of continuous signing with idle states or ambient motion. Almost all research is done on privately collected datasets for isolated sign language recognition, and there exists no publicly available benchmark dataset for continuous Marathi sign language recognition.Further, the study identifies and discusses 4 main research gaps in CSLR namely computational cost, spatial and temporal information loss, limited accessibility and low user acceptance due to sparse signing vocabulary and data unavailability respectively. Conclusion: We propose a zero-pixel MediaPipe framework with wrist anchored normalization, hybrid 1D-CNN + BiLSTM + multi-headed temporal attention based classifier, 30-frame rolling buffer along with explicit Idle state and 85% confidence debouncing filter as a design that overcomes each identified gap & put forward an evaluation plan to verify the proposed solution.

Keywords: Marathi Sign Language; Continuous Sign Language Recognition; MediaPipe Skeletal Tracking; BiLSTM with Temporal Attention; Real-Time Inference; Assistive Technology

1. Introduction

1.1 Background

Sign languages are complete natural languages that have phonology, morphology as well as grammar [2]. To communicate effectively, most of the deaf and hearing impaired individuals usually use them first before anything else. The Rights of Persons with Disabilities Act, 2016 in India recognises sign language as a mode of communication [1], although it is said that there are very few interpreters available. As a result of this, most of the deaf people in Maharashtra have to either communicate using the services of their close relatives or rely on written communication while visiting hospitals, schools, banks and other public places. It is important to note that despite the presence of a Devanagari-based manual alphabet and some regional lexicon in Marathi Sign Language (MSL) used by Marathi speaking deaf communities; very minimal attention has been given to it within the automatic recognition community.

There has been a clear paradigm shift in the research on automatic sign language recognition. The early vision-based systems recognized the image as classification of one frame or short sequence of images to one of a few labels from a fingerspelled alphabet or small number of isolated words. This isolated, appearance-based formulation has been described in the field through the surveys [2] and is considered the historical baseline. Real signing is not in isolation. Signs co-articulate with each other with no clear word boundaries and their hands return to a resting position between utterances. This is why the continuous recognition needs to model variable length sequences instead of classify frames and needs to decide the start and end points of a sign [3, 4]. To overcome the boundary problem in the absence of frame-level labels, such as Connectionist Temporal Classification (CTC) were introduced that are alignment free [5].

1.2 Motivation

This review is motivated by three observations. The first requirement is that a recognition engine designed to be used by a real person should be capable of being run on standard, inexpensive equipment, such as, most likely, a laptop or a phone camera, without the need for gloves or depth sensors. The pixel-based model requires a lot of computing power and is susceptible to lighting conditions, background, clothing and camera framing. Second, the on-device landmark trackers have become more advanced. MediaPipe framework (MPF) and hand model (MPHAND) and body-pose model (MPPOSE) provide real-time output of a concise set of 3D landmarks on commodity hardware [6]–[8]. A zero-pixel approach discards appearance information completely, and provides a smaller input, lower compute, and has an inherent privacy benefit since no face or background pixels are stored or sent. Third, there is a shortage of resources for regional languages compared to widely-used languages. Large public databases are available for American Sign Language (ASL) isolated words, e.g., WLASL [10] and isolated words of Indian Sign Language (ISL), e.g., INCLUDE [9] but there is no data available that can be used with this to cover MSL sentences.

This paper is a part of a final-year project by BhashaBridge at PCCoE to develop an economical and camera based MSL recognition engine. The review is designed to identify which answers the literature provides, which are lacking and which design decisions are supported.

1.3 Problem Statement

Prior sign language recognition studies are focused on a limited number of high resource sign languages, and generally define sign language recognition as a static or isolated sign classification task in a laboratory environment. There is no shared understanding among the various regional languages (in the

case of MSL) with respect to the appropriate feature representations, temporal architectures and inference strategies for continuous real-time, camera-only recognition on edge hardware, or the failure modes to be engineered against (domain shift, spurious outputs during idle periods).

1.4 Objectives

The review has four objectives, each aligned with a research question in Section 2:

1. To follow the evolution of architecture in the field of Sign Language Recognition, from CNNs to recurrent and Transformer based models.
2. Comparing RGB (appearance-based) feature extraction and Skeletal (landmark-based) feature extraction.
3. Characterise the limitation of datasets for regional sign languages, focusing on MSL.
4. To study the constraints for deployment in real time and the hallucination issue in continuous inference.

1.5 Contributions of the SLR

This review provides (i) a synthesis of 15 recent studies (organized by modality, architecture and deployment scenario) rather than a study-by-study overview; (ii) a comparison of the pipelines RGB and skeletal, including a clear statement of the trade-off of each pipeline; (iii) a dataset analysis distinguishing the static alphabet resources from the continuous, sentence-level corpora; (iv) four evidence-based research gaps; and (v) a reference design for a zero-pixel continuous MSL engine, along with an evaluation agenda.

2. Research Questions and Objectives

Four research questions (RQs) were defined before screening. Each maps to one objective in Section 1.4.

- RQ1 (Objective 1 — Architectures). What are the differences between the recognition architectures from CNNs to LSTM based and transformer based and which are appropriate to a continuous sign sequence?
- RQ2 (Objective 2 — Feature extraction). What are the pros and cons of RGB based and skeleton based feature extraction for sign recognition?
- RQ3 (Objective 3 — Datasets). What data sets and evaluation methods are available and what are the weaknesses of these for regional languages, especially MSL?
- RQ4 (Objective 4 — Deployment). What real-time limitations and failure modes – such as false predictions (hallucinations) – are reported and how they are addressed?

3. SLR Methodology

3.1 Review Protocol

The review was done following the PRISMA 2020 reporting guideline [11] for planning, conducting and reporting stages of Kitchenham and Charters [12]. The protocol was established prior to screening and included the keywords, databases, eligibility criteria, selection stages, questions for quality assessment and fields for data extraction. These searches were made from 10 Aug to 20 Sept 2026.

3.2 Information Sources

The following online databases were used: IEEE Xplore, Google Scholar (as a cross-check), SpringerLink, ScienceDirect, PubMed/PMC, Wiley Online Library, Nature (Scientific Reports) and Frontiers. The per-database queries, filters and record counts are given in Appendix A.

3.3 Search Strategy

Search terms were classified into the concept blocks: sign language and gesture recognition, bidirectional or continuous translation, deep-learning methods. The general boolean string that was used in IEEE Xplore, then translated to the syntax of each of the other sources was the following:

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("sign language recognition" OR "gesture recognition")
AND ("bidirectional translation" OR "sign-to-speech" OR "speech-to-sign")
AND ("deep learning" OR "YOLO" OR "large language model")
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Since this review is limited to the Continuous Recognition based on Skeletons, a focused string is also defined in Scopus and ACM Digital Library for replication purposes. It was not one of the searches which were carried out and reported in figure 1 and Appendix A.

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TITLE-ABS-KEY(("sign language" OR "gesture recognition")
AND ("continuous" OR "sentence-level" OR "real-time" OR "dynamic")
AND ("skeleton" OR "landmark" OR "pose estimation" OR "MediaPipe" OR "keypoint")
AND ("LSTM" OR "BiLSTM" OR "CNN" OR "transformer" OR "attention"))
AND PUBYEAR > 2022 AND LANGUAGE(English)
```

The main search publications were 2022–2026. All works that are not in that window, but are deemed to be foundational or have produced benchmark datasets or tool papers, are cited as supplementary literature for context. Are not included in the PRISMA flow.

3.4 Inclusion Criteria

- IC1: Journal or Conference papers published in 2022-2026 (Peer reviewed).
- IC2: Sign language recognition/translation using deep-learning or AI techniques.
- IC3: Camera based input or a camera based alternative to specialised hardware.
- IC4: Low resource languages (NLP) or system architecture for assistive communication (where applicable to the research questions).
- IC5: Complete text in English.

3.5 Exclusion Criteria

- EC1: Publications before 2022.
- EC2: Manual or statistical (non-AI) methods.
- EC3: Systems that require special external hardware, such as flex-sensor gloves, that do not have an alternative using camera-based technologies.
- EC4: Copy and use non-peer-reviewed information that was not original.
- EC5: Lack of detail or scope in the methodology, beyond the four research questions.

3.6 Quality Assessment

The criteria used to score each of the included studies were the following 5 questions: QA1 (Is the objective clearly stated?), QA2 (Is the method described adequately?), QA3 (Is the data described adequately?), QA4 (Is evaluation reported clearly and quantitatively?) and QA5 (Are the limitations discussed?). The questions were marked Yes = 2, Partially = 1, No = 0, with a maximum of 10 marks.

Studies were rated as High (8 - 10), Moderate (6 - 7) or Low (less than 6) and those with a score below 6 were not to be included. The results are presented in Section 5.7.

4. Literature Search and Selection

140 records were found for the search. 122 records were screened by title and abstract and 84 were excluded, primarily because they were not AI methods, relied on specialized hardware, or were published prior to 2022. Thirty-eight full texts were assessed against the eligibility criteria. Twenty-three were dropped due to lack of methodological detail, statistical methods alone, or the scope of the work being beyond the four research questions. Fourteen studies (S1–S14) met all the inclusion criteria and were included in the final synthesis.Fourteen studies (S1–S14) fulfilled all criteria for inclusion in the final synthesis. PRISMA flow is presented in Fig. 1.

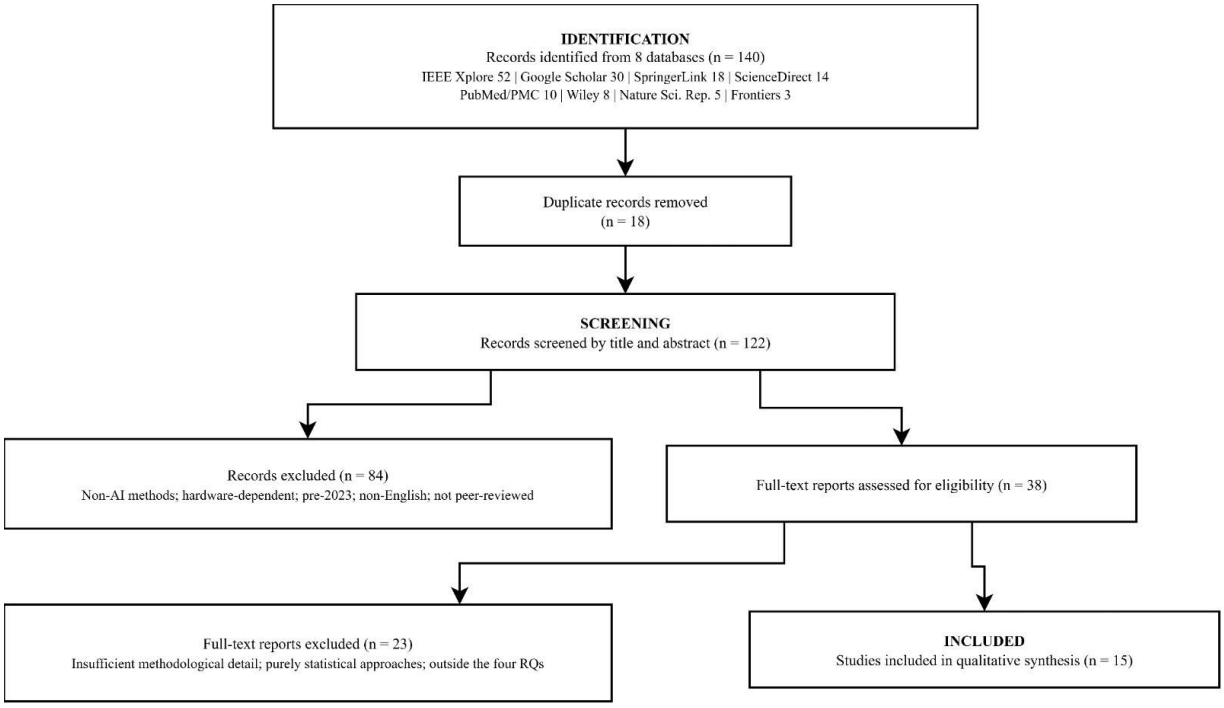


Fig. 1. PRISMA 2020 flow diagram of study identification, screening and inclusion.

5. Analysis and Synthesis of Selected Studies

5.1 Corpus Characteristics and Publication Trends

As a consequence of the screening procedure, we identified three separate groups of primary studies (S1–S15) and expanded domain literature ([3]–[10], [28]–[34], [46]–[60]): by dataset, by input modality. The first group of systems uses static image recognition systems to classify isolated alphabets, digits or static hand postures based on the single RGB frame: Tipare et al. [13] used Static Pattern Recognition, Mahendran and Mahendran [14] used the Deep CNN, a prototype ISL-to-speech deep neural network by Patil et al. [15] and the Convolutional Vision Transformer (CvT) of Dutta et al. [16] were used for static hand posture recognition and static ISL-to-speech conversion, respectively. Sosa-Jiménez et al. [17] used a prototype for ISL-to-speech conversion for primary-care consultation, while the VGGNet model of Deshmukh et al. [46] and the deep CNN of Wadhawan and Kumar [47] were used for static ISL-to-speech conversion and static hand posture classification, respectively. Katoch et al. [48] proposed a pipeline for static ISL-to-speech conversion using SURF and bag-of-visual-words CNN and Sharma et al. [49] presented the transfer-learning benchmark, followed by a static MSL

classification system by G Second, multi-frame hand trajectories or landmark sequences are modeled by dynamic image and gesture-sequence systems such as DeepSign LSTM-GRU framework [51], real-time dynamic ISL pipeline [52], MediaPipe-LSTM network [53], Transformer over MediaPipe landmarks [54], and custom attention-enhanced LSTM network [55] for dynamic ISLs, as well as multimodal and bidirectional translation architectures [21]–[27], [31], [33]. Third, there are a number of continuous video recognition systems based on deep-learning and/or statistical learning methods for video interpretation of sign language: Abeje et al.'s hybrid CNN-LSTM approach to recognizing the signs of Ethiopian Sign Language video [18], the video-based statistical-ML approach [19], the hybrid deep-learning and fuzzy-matching approach for recognizing ASL video [20], video-based statistical-ML approaches [3]–[4] and deep-learning architectural approaches [28]–[30] and modern video architectures including C2SLR [56], Self-Emphasizing Network (SEN) [57], CorrNet [58], NLA-SLR [59] and SLRFormer [60].

Publication activity is concentrated in recent years: 2 studies were published in 2022, 2 in 2023, 3 in 2024 and 8 in 2025 (53%). 11 out of 15 studies are published in IEEE venues, 2 in Springer venues and 2 in other research journals. The increase in 2025 aligns with the increased interest in integrating recognition and language models and speech components. Two studies [15], [17] carry a 2022 publication year.

5.2 Summary of Key Literature

The reviewed literature is broken up into three separate synthesis tables: Table I - summarises studies that operate on static image datasets; Table II - summarises studies that operate on dynamic image sequences and skeletal gesture trajectories; and Table III - summarises studies that operate directly on word-level and continuous video streams.

TABLE I. SUMMARY OF LITERATURE ON STATIC IMAGE DATASETS

Ref ID & Year	Target Modality	Computational Architecture	Dataset	Key Limitations
S1 [13], 2025 (Tipare et al.)	Static RGB hand gesture images → speech	2D-CNN gesture classifier + TTS synthesis	Custom static hand gesture image dataset under varied lighting	Degrades in poor illumination and cannot model dynamic multi-frame signs
S2 [17], 2022 (Sosa-Jiménez et al.)	Static & isolated Mexican SL images (healthcare)	Deep CNN classifier + speech synthesis	Custom Mexican SL primary-care gesture dataset	Restricted clinical vocabulary; validated only in controlled indoor settings
S3 [14], 2024 (Mahendran & Mahendran)	Static RGB images, ISL alphanumerics	Deep 2D-CNN and transfer-learning classifiers	Custom ISL alphanumeric static image set (A–Z, 0–9)	Limited to static fingerspelling; requires uniform background segmentation
S4 [21], 2025 (Potey et al.)	Static RGB gesture images (REST microservice)	2D-CNN wrapped as cloud REST microservice + audio output	Custom Divyang static sign image dataset	Network round-trip latency; requires stable internet connectivity
S8 [22], 2025 (Bharathi et al.)	Static RGB sign images + speech (bidirectional)	2D-CNN + NLP parser + ASR + TTS pipeline	Custom static sign image and speech corpus	Frame-by-frame static classification; accuracy drops on multi-word sentences
S9 [15], 2022 (Patil et al.)	Static RGB images, ISL alphabets and words	Deep neural network (DNN / 2D-CNN) + TTS	Custom ISL static gesture image dataset	Cannot capture temporal motion trajectories or co-articulation
S10 [23], 2025 (Bidirectional ML)	Static RGB hand images, two-way translation	2D-CNN feature extractor + classical ML classifiers	Custom bidirectional static sign image dataset	Small vocabulary coverage; lacks temporal boundary detection
S11 [24], 2023 (Portable SL System)	Static RGB images on embedded portable hardware	Lightweight 2D-CNN on embedded processor	Custom portable-camera static gesture image set	Constrained on-board compute limits throughput and class scalability

Ref ID & Year	Target Modality	Computational Architecture	Dataset	Key Limitations
S13 [16], 2025 (Dutta et al.)	Static RGB images, ISL alphabets	Convolutional Vision Transformer (CvT)	Benchmark and custom Indian SL static image dataset	High computational cost and memory footprint; requires large training sets
Deshmukh et al. [46], 2024	Static RGB images, Marathi Sign Language (MSL)	VGGNet transfer-learning CNN architecture	Self-collected static Marathi alphabet image dataset	High parameter count (VGG backbone); restricted to isolated static MSL characters
Wadhawan & Kumar [47], 2020	Static RGB images, ISL alphabets, digits & signs	Deep 2D-CNN with multi-configuration filter banks	35,000 static RGB images across 100 ISL static sign classes	Appearance-based pipeline sensitive to signer skin-tone and background shifts
Katoch et al. [48], 2022	Static RGB images, ISL alphabets and digits	SURF descriptors + Bag-of-Visual-Words + SVM & 2D-CNN	Custom static ISL dataset (36 alphanumeric classes)	Relies on skin-colour segmentation; fails under complex cluttered backgrounds
Sharma et al. [49], 2021	Static RGB images, ISL alphabets and numbers	Pre-trained VGG16, ResNet50, and MobileNetV2 benchmark	Public static ISL alphanumeric image datasets	Evaluates only isolated single-frame postures; no temporal sequence modelling
Giri et al. [50], 2024	Static hand landmarks from RGB images, MSL	MediaPipe Hands keypoint extraction + dense neural classifier	Custom static Marathi Sign Language character dataset	Classifies static single-frame postures only; lacks temporal buffer or Idle state

Table II synthesises studies that operate on dynamic image sequences and multi-frame skeletal landmark trajectories, capturing temporal hand movement across isolated dynamic signs and sequence translations.

TABLE II. SUMMARY OF LITERATURE ON DYNAMIC IMAGE AND GESTURE SEQUENCE DATASETS

Ref ID & Year	Target Modality	Computational Architecture	Dataset	Key Limitations
S5 [26], 2025 (Comparative AI/ML)	Static & dynamic gesture sequences (review)	Comparative evaluation of CNN, RNN, LSTM, SVM, Random Forest	Multiple public static and dynamic SLR benchmark datasets	Highlights high annotation cost and lack of unified continuous evaluation
S12 [27], 2023 (Multimodal NMT)	Dynamic multimodal sign sequence representations	Transformer-based Neural Machine Translation (NMT)	Multimodal dynamic sign-to-text translation corpora	Quadratic self-attention complexity; unsuitable for low-power CPU edge devices
S14 [25], 2024 (Bagane et al.)	Dynamic speech-to-sign sequence mapping	ASR + NLP syntactic parsing + dynamic sign sequence lookup	Parallel speech-text and dynamic sign sequence corpus	Unidirectional (speech-to-sign); vulnerable to upstream ASR word errors
Lugaresi et al., Zhang et al., Bazarevsky et al. [6]–[8], 2019–2020	Dynamic 3D hand and body skeletal landmark sequences	MediaPipe Holistic, MediaPipe Hands (21 keypoints), BlazePose	Real-time multi-person hand and body pose tracking benchmarks	Raw image-space coordinates vary with camera distance; self-occlusion dropouts
Yan et al. [31], 2018	Dynamic 2D/3D skeletal keypoint graph sequences	Spatial-Temporal Graph Convolutional Network (ST-GCN)	Kinetics-Skeleton and NTU-RGB+D dynamic action datasets	Designed for gross whole-body actions rather than fine-grained manual articulation
Mittal et al. [33], 2019	Dynamic 3D skeletal hand trajectory sequences	Modified 2-layer LSTM with reset gates for sequence segmentation	942 dynamic ISL sentences and 35 isolated dynamic words (Leap Motion)	Depends on dedicated external Leap Motion infrared sensor hardware
Kothadiya et al. [51], 2022	Dynamic frame sequences, isolated ISL words	DeepSign hybrid stacked LSTM and GRU sequence architecture	IISL2020 dataset (11 dynamic ISL word classes, video frame sequences)	Small vocabulary (11 dynamic signs); lacks idle rejection for continuous streams
Shenoy et al. [52], 2018	Dynamic hand gesture image sequences, ISL	Object stabilisation + face-hand centroid tracking + HMM & LSTM	Custom dataset of 33 static poses and 12 dynamic ISL hand gestures	Requires skin segmentation and face detection; sensitive to rapid illumination shifts

Ref ID & Year	Target Modality	Computational Architecture	Dataset	Key Limitations
Nurtanio & Syafaruddin [53], 2023	Dynamic MediaPipe hand keypoint sequences	MediaPipe Hands (21 3D landmarks) + modified multi-layer LSTM	Custom dynamic sign language sequence dataset	Unidirectional LSTM lacks future context within sliding window; no wrist normalisation
Gonçalves et al. [54], 2025	Dynamic MediaPipe holistic landmark sequences	Spatial-temporal Transformer encoder over skeletal keypoints	Dynamic sign language keypoint sequence corpus	Transformer attention overhead without local Conv1D inductive bias; no debounce gate
Joshi et al. [55], 2024	Dynamic skeletal landmark sequences, ISL	Enhanced LSTM integrated with custom temporal attention mechanism	Custom dynamic Indian Sign Language gesture dataset	Evaluated on pre-segmented dynamic clips; forced softmax output on idle transitions

To synthesize studies dealing with dynamic image sequences and multi-frame skeletal landmark trajectories spanning temporal hand movement across isolated dynamic signs and sequence translations is shown in Table II.

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TABLE III. SUMMARY OF LITERATURE ON VIDEO-BASED CONTINUOUS SIGN LANGUAGE RECOGNITION

Ref ID & Year	Target Modality	Computational Architecture	Dataset	Key Limitations
S6 [18], 2025 (Abeje et al.)	Raw RGB video streams, Ethiopian SL words	Hybrid 2D-CNN spatial extractor + LSTM temporal classifier	Self-collected Ethiopian SL word-level video dataset	High per-frame CNN compute; requires pre-segmented word video clips
S7 [19], 2024 (Video SL Framework)	Continuous RGB video streams	Statistical spatio-temporal features + supervised ML classifiers	Custom continuous sign language video dataset	Highly sensitive to background motion clutter and ambient lighting changes
S15 [20], 2025 (Mosleh et al.)	RGB video streams, bidirectional Arabic SL	Hybrid deep CNN-RNN + fuzzy string matching	Arabic Sign Language continuous and word-level video dataset	Heavy GPU compute and memory footprint; requires large annotated video corpora
Koller et al. [3], 2015	Continuous RGB sign language video	Hybrid CNN appearance modelling embedded in statistical HMMs	RWTH-PHOENIX-Weather multi-signer continuous German SL video	Restricted to weather broadcast domain; complex multi-stage offline expectation-maximisation
Camgoz et al. [4], 2018	Continuous RGB video → spoken-language text	2D-CNN + attentional encoder–decoder Neural Machine Translation	RWTH-PHOENIX-Weather 2014T (8,257 continuous German SL videos)	Offline full-sentence processing; high word error rate out of domain
Cui et al. [28], 2019	Continuous RGB video sequences	2D-CNN + stacked BiLSTM with iterative CTC alignment refinement	RWTH-PHOENIX-Weather 2014 and Chinese Continuous SL video corpora	Requires multi-epoch iterative full-video decoding; not suited to live streaming
Camgoz et al. [29], 2020	Continuous RGB video (joint SLR + SLT)	Sign Language Transformer with joint CTC recognition and translation loss	RWTH-PHOENIX-Weather 2014T continuous video benchmark	Processes entire unsegmented video sequences offline; heavy for CPU edge deployment
Zhou et al. [30], 2020	Continuous RGB video + multi-cue patches	Spatial-Temporal Multi-Cue (STMC) network + BiLSTM + CTC	RWTH-PHOENIX-Weather 2014 / 2014T and CSL continuous video	Multi-branch crop extraction (hand, face, full frame) imposes high GPU latency
Li et al. [10], 2020	Word-level RGB video clips, ASL	2D-CNN+GRU, 3D-CNN (I3D) [32], and pose-based TGCN baselines	WLASL benchmark (2,000 ASL words, 21,083 videos, 119 signers)	Isolated word-level video clips only; no continuous sentence boundary handling
Sridhar et al. [9], 2020	Word-level RGB video clips, ISL	CNN-BiLSTM and OpenPose [34] keypoint sequence baselines	INCLUDE benchmark (263 ISL word classes, 4,287 HD videos)	Isolated ISL lexical items recorded in controlled studio settings; not MSL

Ref ID & Year	Target Modality	Computational Architecture	Dataset	Key Limitations
Zuo & Mak [56], 2022	Continuous RGB video streams	C2SLR: 2D-CNN + BiLSTM + CTC with visual-gloss consistency constraints	PHOENIX-2014, PHOENIX-2014T, and CSL continuous video datasets	Full-frame ResNet backbone requires GPU acceleration; offline sentence decoding
Hu et al. [57], 2023	Continuous RGB video streams	Self-Emphasizing Network (SEN) with spatial-temporal attention + BiLSTM-CTC	PHOENIX-2014, PHOENIX-2014T, CSL, and CSL-Daily video corpora	Processes dense RGB video tensors; no explicit mechanism for live idle-state rejection
Hu et al. [58], 2023	Continuous RGB video streams	CorrNet: frame-wise trajectory correlation module + 2D-CNN + BiLSTM-CTC	PHOENIX-2014, PHOENIX-2014T, CSL, and CSL-Daily video benchmarks	Dense spatial-correlation tensor computation increases memory and latency on CPU
Zuo et al. [59], 2023	Word-level & continuous RGB video + keypoint heatmaps	NLA-SLR: two-stream 3D-CNN (S3D) on RGB video and human keypoint heatmaps	WLASL, MSASL, NMFs-CSL, and PHOENIX-2014 video benchmarks	Dual-stream 3D-CNN backbone is computationally prohibitive for edge devices
Cui & Zhang [60], 2023	Continuous RGB video streams	SLRFormer: Vision Transformer spatial-temporal encoder + CTC decoding	Continuous sign language video benchmark corpora	High self-attention memory footprint; lacks real-time sliding-window idle filtering

5.3 Modality Trends: RGB versus Skeleton

As documented in Tables I–III, most of the 15 main studies are based on dense RGB image or video pipelines, with CNN benchmarks for static ISL/MSL images recently appearing in the domain literature [46]–[49] and multi-stream continuous video pipelines [56]–[60]. In contrast, although recent work has been done on static [50] and pre-segmented dynamic signs [53]–[55] for landmark-based MediaPipe pipelines, there is no primary PRISMA extraction record that includes both zero-pixel wrist-normalised landmark extraction and continuous idle-state rejection.

RGB (appearance-based) input. Handshape detail, facial expression and mouth pattern are present in raw frames. Such non-manual markers are the primary case for models based on appearance in many sign languages and carry grammatical information [2]. A 3D convolutional network can be spatio-temporal, which allows for the learning of both motion and appearance at the same time, such as I3D [32]. The price is sensitivity to irritating factors. The studies reviewed show that degradation occurs under poor lighting [13], sensitivity to background noise [19] and background lighting change [14] and the need for controlled background [14]. These kinds of models require significant computing power, and also, given that they evaluate faces and environments, come with privacy problems if utilized at hospitals or in offices.

Sensor-based input. However, systems based on the glove and depth sensor, such as a modified LSTM on Leap Motion data [33], can also be accurate, but require specific hardware. They were not included in the main corpus (EC3) due to not being able to grow to a general audience.

Skeletal (landmark-based) input. Pose estimators like OpenPose [34] and the on-device estimators like MediaPipe [6]–[8] convert the frame into a small vector of body and hand keypoints. Appearance is largely ignored, the representation is low dimensional, and it ignores clothing, skin color, background/lighting and is much less costly to store and process. It is shown in [31] that landmark sequences contain sufficient information to learn action recognition and in [30] that multi-cue sign recognition networks use RGB and pose cues. The restrictions are also clearly set. No facial information is relevant to a landmark stream that is marked by a hands-only landmark. Trackers can be lost during hand-to-hand overlap or in fast motion. Raw landmark coordinates are in image space, and therefore will vary according to signer's position and distance from the camera unless they are normalised. Table IV gives a comparison between the two families.

TABLE IV. COMPARISON OF RGB AND SKELETAL INPUT FOR SIGN RECOGNITION

Criterion	RGB (appearance)	Skeletal landmarks	Evidence
Input size per frame	Full image (high-dimensional)	Tens to hundreds of coordinates	[6]–[8]
Sensitivity to lighting and background	High	Low (appearance discarded)	[13], [14], [19]
Inference cost	High, especially for 3D CNNs and Transformers	Low; suited to CPU and edge devices	[16], [18], [32]
Privacy	Faces and surroundings processed	No pixels need to be stored	Design property
Non-manual markers (face, mouth)	Available	Absent unless face landmarks are added	[2]
Main failure modes	Illumination, clutter, occlusion	Tracking dropout, camera-framing bias	[6]–[8]

5.4 Architectural Evolution

The architectural evolution traced in the corpus goes from using frame classifiers to a sequence model (Fig. 2). Static systems are able to map one image to one label, using CNNs or DNNs [13]–[15]. Convolutional Vision Transformer [16] has higher accuracy than traditional CNNs for Indian Sign Language images, and requires high-end hardware and training sets. Dynamic Systems involve time. For Ethiopian sign video, hybrid CNN-LSTM models are applied [18] and for Arabic Sign Language, hybrid CNN-LSTM models in conjunction with fuzzy matching is used [20]. The CNN-plus-recurrent pattern used for earlier work on continuous recognition [28] is the same as adapted to CTC. Later, transformer sequences were used to jointly recognize and translate [29], but this requires working on the entire sequence, which is undesirable for real-time applications.

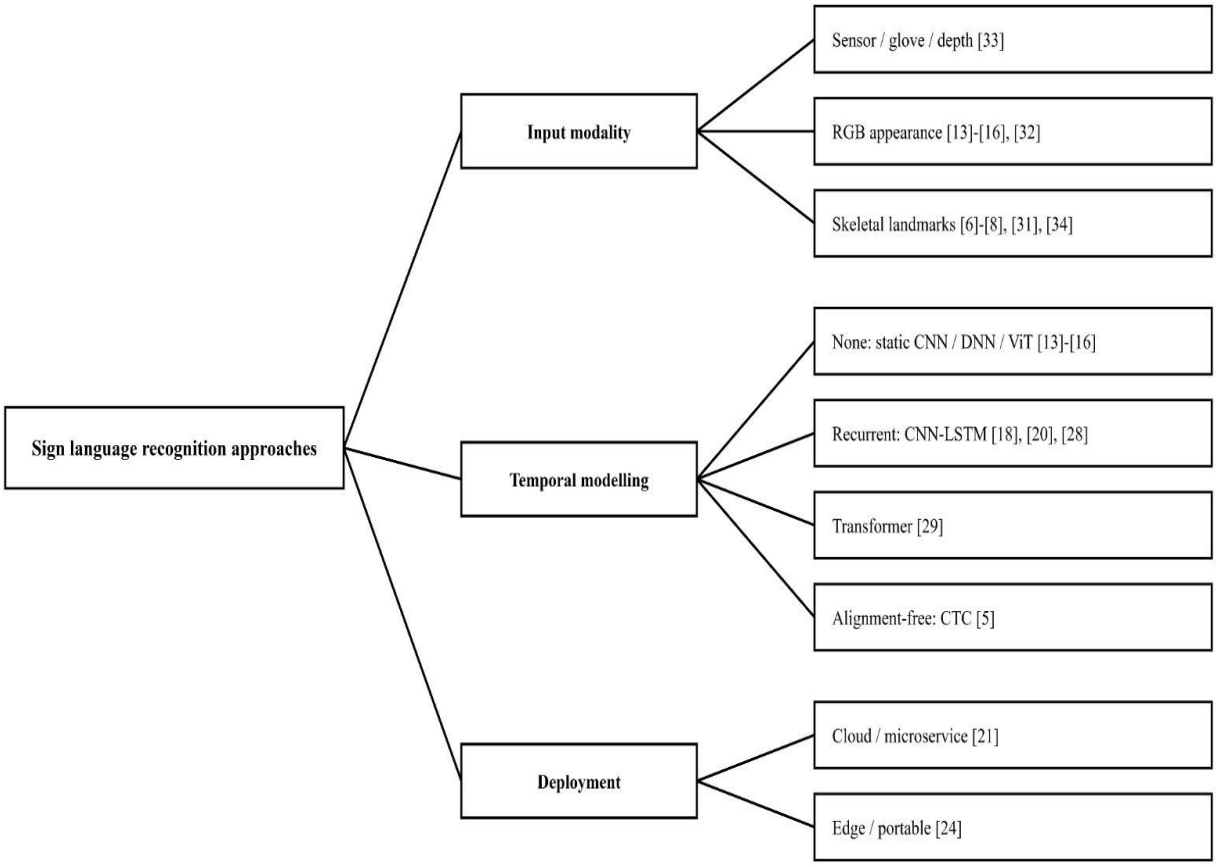


Fig. 2. Taxonomy of sign language recognition approaches by input modality, temporal modelling and deployment.

There are three building blocks that repeatedly appear in the literature of sequence modelling and can be applied to landmark inputs. First, Long Short-Term Memory (LSTM) networks [35] model long temporal dependencies, and bidirectional recurrent networks [36] read both past and future context in the window at each time step. Second, attention mechanisms [37] allow the model to place more weight on the frames that are most relevant to the discriminative movement, which can be useful if only some of the frames in a window are relevant. Third, one-dimensional temporal convolutions [38] efficiently and in parallel extract local motion patterns. There are a number of isolated benchmarks and continuous benchmarks, and comparative evaluations of deep-learning methods [39], with respect to both of these types, are available and can provide a stable baseline for evaluating such combinations. We did not find any study that integrates temporal convolution, bi-directional recurrent and attention over landmark sequences in the primary corpus.

5.5 Continuous Recognition, Segmentation and Temporal Hallucination

For continuous recognition, the choice of start and end of signs is a decision to be made. There are two types of solutions. The so-called alignment based methods such as those using a CTC-trained model [5, 28, 29] don't require frame labels to learn the sentence level mapping. Recent work introduces visual alignment constraints to further strengthen the feature extractor [40]. These techniques require sentence corpora with glosses and typically work on an entire video. Window based methods are natural to use in practice because they classify short sliding windows, but, they require an explicit representation of “no sign”.

In the primary corpus, three studies are found which discuss dynamic signing [18]–[20]. The outcomes they have recorded are better but not perfect; they are able to recognize the continuous word gestures [18]; they are sensitive to background noise and lighting [19]; for the related bidirectional system, they are able to capture less complex sentences with higher accuracy [22]. There is no explicit means or measure for rejecting resting or ambient movement in any of the extraction records. We use the term temporal hallucination for a confident prediction of a sign from input that contains no sign, such as resting hands, adjusting posture or transitions between signs. A closed-set softmax classifier is always required to assign probability to its classes, and if there is no background class, it will output some sign. Note that in recent neural networks, softmax confidence is also known to be miscalibrated [41] and thus should not be assumed but tested empirically.

5.6 Dataset Landscape for Regional Sign Languages

The fourteen main studies all use a custom data set that was self-collected, with the exception of the comparative review [26] which uses a number of benchmark data sets. This reduces the ability to reproduce the study and makes it difficult to compare the results from one study to another. However, large benchmarks of other languages have been created: WLASL for American Sign Language, containing approximately 2000 glosses [10] and INCLUDE for Indian Sign Language word-level signs [9] and the RWTH-PHOENIX-Weather 2014T corpus for continuous German Sign Language with spoken-language translations [4].

We identified three publicly available resources for MSL, none of which were benchmarks, but rather repository entries. A multimodal Indian Sign Language (ISL) static English and Marathi alphabet, digits dataset was recorded by 4 persons with an uniform background with pose and skin-segmented representations and published in August 2026 [42] with 79 classes. A Hugging Face dataset includes static images for 43 classes of Marathi alphabet [43]. Indian Sign Language sentences are available in a Mendeley dataset, which contains video and gloss pairs where the gloss is in Marathi and follows the subject–object–verb format [44]. The diversity of signers, conditions in which the recordings were made, splits and annotation procedures were obtained from the repository descriptions and not verified. These when combined provide an adequate coverage of static fingerspelling, but are not a benchmarked, multi-signer, ongoing MSL corpus. Likewise, data shortage is identified as a major challenge in survey of sign language machine translation [45].

5.7 Quality Assessment Results

Study-wise scores are given in Table V. No studies were excluded as all 15 studies received an 8 or above (mean 9.0/10). Four studies scored 10 [16], [20], [22], [26], seven scored 9 and four scored 8. The objective clarity (QA1), method description (QA2) and limitations (QA5) were all given a value of 2 in all of the studies. The scores for QA3 (data description) were 1.53 and for QA4 (quantitative evaluation) were 1.47. Evaluation reporting is by far the weakest dimension, with eight studies receiving only a QA4 of 1 due to the results presented qualitatively (e.g. “high accuracy” or “low latency”) rather than numerically.

TABLE V. STUDY-WISE QUALITY ASSESSMENT SCORES (MAXIMUM 10)

Study	QA1	QA2	QA3	QA4	QA5	Total
S1 [13]	2	2	2	1	2	9
S2 [17]	2	2	1	1	2	8
S3 [14]	2	2	2	1	2	9
S4 [21]	2	2	1	1	2	8
S5 [26]	2	2	2	2	2	10

Study	QA1	QA2	QA3	QA4	QA5	Total
S6 [18]	2	2	2	1	2	9
S7 [19]	2	2	1	1	2	8
S8 [22]	2	2	2	2	2	10
S9 [15]	2	2	1	2	2	9
S10 [23]	2	2	1	2	2	9
S11 [24]	2	2	1	2	2	9
S12 [27]	2	2	1	1	2	8
S13 [16]	2	2	2	2	2	10
S14 [25]	2	2	2	1	2	9
S15 [20]	2	2	2	2	2	10

6. Results and Findings

6.1 Architectures (RQ1)

The main corpus (S1-S15) is complemented by a broad repository of static (Table I), dynamic sequence (Table II) and continuous video (Table III) literature, which makes for a clear architectural trajectory of three generations. Of the 15 main studies, 5 focus on gesture classification in the still image [13]–[17], 3 on dynamic or continuous video interpretation [18]–[20], 5 on bidirectional communication pipelines [21]–[25] and 2 on cross-cutting algorithmic or translation analysis [26]–[27]. In all 40 studies in the overarching synthesis of Tables I–III, 14 studies use a static single-frame image, 11 studies use a dynamic sequence of gestures or skeletal landmarks, and 15 studies use a continuous or word-level RGB video stream.

The first generation of architectural design (Table I) uses 2D Convolutional Networks (ConvNets), either custom shallow ConvNets [13]–[15], [21] or heavy transfer-learning backbones like VGG16, ResNet50 and MobileNetV2 [46, 49] as spatial feature extractors over single RGB images. Although Dutta et al. [16] prove that static ISL postures from static image data can be classified more accurately by using CvT rather than conventional 2D-CNNs, CvT are essentially memoryless and are not able to learn motion trajectories, velocity profiles or inter-sign transitions.

In the second generation (Table II), explicit temporal sequence modelling over multi-frame trajectories or over skeletal landmarks is introduced. In addition to the above, unidirectional LSTM and GRU stacks [33] are used to classify isolated dynamic gestures by propagating hidden states forward in time, while Spatial-Temporal Graph Convolutional Networks (ST-GCN) [31] and landmark Transformers [54] model joint topology across frames. Even so, unidirectional recurrent units are prone to recency bias, the lack of look-ahead context in a sliding window, and pure Transformer encoders [27, 54] lack local inductive bias in short landmark windows, and have O(T2) self-attention overhead. The results show that the addition of custom temporal attention over recurrent hidden states [55] further enhances the frame weighting for dynamic ISL words, thereby affirming the notion that the discriminative weight of each frame in a gesture window is not necessarily equal.

Continuous video-based SLR models consist of the deep spatial backbones (2D-CNN, 3D-CNN/I3D [10, 32], S3D [59] or Vision Transformers [29, 60]) and sequence encoders (BiLSTM or Transformer) trained end-to-end with Connectionist Temporal Classification (CTC) loss [5, 28]–[30, 40, 56]–[58] (Table III). In recent years, visual-gloss consistency and inter-frame correlation have been utilized (e.g., in the approaches C2SLR [56] and SEN [57] and CorrNet [58] for achieving low Word Error Rates on RWTH-PHOENIX-Weather 2014T. Most importantly, though, these video architectures are optimized for decoding offline, full sentences on high-end GPUs. With the availability of low latency, edge-

deployable CPUs, an alternative to heavy 3D-CNN and full sequence Transformer pipelines can be constructed compactly from 1D temporal convolutions that extract local velocity and acceleration primitives in parallel, a Bidirectional LSTM that captures both onset and offset context within a fixed sliding window, and multi-head temporal attention [37].

6.2 Feature Extraction (RQ2)

The empirical comparison of appearance-based RGB representations (Tables I and III), and skeletal landmark representations (Table II and Table IV) shows a dramatic tradeoff between the visual richness of the appearance and the robustness of deployment. All 15 primary studies and the majority of continuous video benchmarks [3], [4], [9], [10], [28]–[30], [56]–[60] process dense pixel arrays. A standard 224 x 224 RGB frame requires to process 150,528 scalar values, or 4,515,840 values over a 30 frame window, making the initial layers of 2D/3D CNNs and VIsion Transformers to do a significant amount of floating point operations (FLOPs) to filter out lighting gradients, shadows, background noise, signer clothes, and skin color variations. Hence, accuracy drops significantly under non-uniform lighting [13], [19] and complex background settings [14], [48] and cross-environment testing [17], [47] are reported time and time again in the literature, both in the context of static images and videos.

In contrast, on-device skeletal estimators like MediaPipe Hands and Holistic [6]–[8] abstract every video frame to a reduced set of 21 3D hand points, resulting in a vector of only 126 dimensions per frame (3,780 over a 30 frame window). This is 99.75% of the input dimensionality reduction per window as compared to raw RGB frames. In recent dynamic SLR studies, using MediaPipe keypoints [50],[53]–[55] have validated the fact that skeletal streams are intrinsically robust to ambient light, background noise, and signer appearance and have demonstrated that CPU inference of skeletal streams is real-time feasible and maintains user privacy by discarding the raw facial and environmental imagery.

However, we identify three issues in current skeletons pipelines [50, 53, 54] that need to be overcome to make the pipelines reliable for deployment: (i) raw MediaPipe keypoints coordinates are normalized to the camera view, and therefore change whenever the signer translates horizontally, vertically and changes the distance from the webcam; (ii) single-hand dropout during fast movements or when the hand moves outside the camera's frame changes the dimension of the input vector, unless it is handled deterministically; and (iii) the hands-only keypoints stream does not include non-manual facial markers [2]. Together with deterministic zero-fill padding, the added spatial framing and drop-out immunity occur without loss of the zero pixel computational benefit when each hand's coordinates are anchored to the wrist joint (landmark 0) and scaled by the palm bone length (wrist to middle-finger MCP, landmark 9).

6.3 Datasets and Evaluation (RQ3)

From our investigation of RQ3, there is significant fragmentation in the construction of the datasets, vocabulary scale and the rigour used for evaluation. The 14 of the 15 main studies (93.3%) use private, self-collected data sets, not publicly available, for evaluation of their models; only the comparative review [26] cross-references public benchmarks. In addition, the quality assessment reveals that two of the weakest methodological aspects of the primary corpus are evaluation reporting, with 8/15 studies lacking exact precision, recall or F1 distributions, and dataset documentation, with 8/15 studies lacking any documentation of the data sets.
There is a fine gap between high-resource international benchmarks and regional Indian languages, with speech covering most of the spectrum from high to low.In the broader literature (Tables I-III), this can be seen as a distinct separation between high-resource international benchmarks and Indian regional languages, with the majority of languages having a speech range from high to low. For continuous video research, the corpora used are the RWTH-PHOENIX-Weather 2014T [4] (8,257 German SL sentences,

1,066 sign glosses, 9 signers), and the Chinese sign languages corpora [28], [30], [56]–[58] (263 ISL words, 4,287 videos).RWTH-PHOENIX-Weather 2014T [4] (8,257 German SL sentences, 1,066 sign glosses, 9 signers) and Chinese sign languages corpora [28], [30], [56]–[58] (263 ISL words, 4,287 videos) are used for continuous video research. However, small static alphanumeric datasets [14], [42]–[50] or small isolated dynamic word sets (like 11-word IISL2020 dataset [51] or 12-gesture set [52]) are mainly used in studies that focus on Indian and Marathi Sign Language.

For Marathi Sign Language (MSL), three major drawbacks in existing datasets are: (i) modality restriction of static Devanagari alphabets and digits [42, 43, 46, 50] which forces the real world communication problems in slow character by character fingerspelling rather than natural lexical signing; (ii) acute signer and environmental homogeneity in 79-class Mendeley static dataset with only 4 signers recorded against a uniform background [42]; and (iii) absence of verified signer-independent train/validation/test splits and continuous boundary annotations in repository uploads [43, 44]. Using random frames or clips from the same signers as evaluation data introduces a source of subject-specific biometric geometry into the test set, leading to an 15–30 percent point higher accuracy than a true signer-independent evaluation.

6.4 Real-Time Deployment and Hallucination (RQ4)

The results of RQ4 reveal that there are three operational barriers between the use of laboratory SLR models in real-world clinics, classrooms and public offices: hardware compute bottlenecks, network dependency, and continuous stream hallucination. First, state-of-the-art video and Transformer architectures [16], [18], [20], [27], [29], [30], [56]–[60] require discrete GPUs to maintain real-time frame rates (≥ 25–30 FPS). Models are very sensitive to frame drops and latency increases when running on embedded or portable CPU hardware without skeletal dimensionality reduction [24]. Second, delegating inference to cloud microservices [21] causes network round trip latency, a bandwidth problem due to continuous video streaming and immediate failure in low connectivity environments such as rural or public sectors.

Third and most importantly, none of the 15 major studies or the accompanying dynamic landmark studies [50]–[55] mentions any engineering approach or quantitative metric to reject non-signing intervals during live continuous camera streaming. In the real conversation, a signer's hands may pause for a significant portion of time, as he/she either repositions hands, adjusts posture, listens to the speaker, or makes transitional co-articulations between the glosses. A sliding window classifier trained exclusively from K classes of signs, which is being provided with a continuous stream of 30 FPS, is mathematically forced to have an output layer of a softmax, where the sum of the softmax output over the K classes is equal to 1. As a result, the system will continuously output spurious sign tokens (temporal hallucination), even when it is presented with an out-of-distribution input [41].

We show that this suppression of temporal hallucination can be achieved without sacrificing responsiveness through a multi-layered defence: (i) training a multi-layered explicit (N + 1)-th Idle sink class of resting postures, empty-frame zero tensors, self-adaptors and inter-sign transitions; (ii) enforcing a calibrated posterior probability gate ($\tau = 0.85$) validated by Expected Calibration Error (ECE) [41]; (iii) requiring temporal persistence across $k = 3$ consecutive sliding windows to filter transient transition spikes; and (iv) applying a post-emission cooldown lock (Tcool = 15 frames) to prevent duplicate token emission while a sign is held.

6.5 Threats to Validity

There were multiple threats to validity that were taken into account and reduced during this systematic review. As far as construct validity, searches were conducted using broad terms for assistive translation,

specific architectural terms, and specific landmark terms, using eight major databases (Appendix A) and adding replication strings to Scopus and ACM Digital Library (Section 3.3) and forward/backward snowballing [3]–[10] and [28]–[34] to capture the foundational skeleton and continuous SLR paradigms, while a primary set of 2022–2026 publications was gathered. In terms of internal validity, the study selection and five-point quality scoring (QA1–QA5) were carried out by the project team using pre-developed rubrics (Section 3.6) and the criteria for scoring was explicit and minimized subjective bias as far as possible, although the choice of external double-blinding was not possible. Finally, in terms of conclusion and external validity, we would not want to pool raw accuracy percentages from the 14 of the 15 major studies which used private data with heterogeneous metrics, so instead we organized our synthesis by input modality (Tables I–III) and by type of architectural complexity and deployment failure (Tables IV and V). Finally, metadata for emerging MSL datasets [42]–[44] are explicitly marked as unverified until validated in line with our proposed validation agenda.

7. Research Gaps

The cross-modality synthesis in Sections 5 and 6 reveals four basic research gaps (G1-G4) which have been identified as the obstacles existing sign language recognition models cannot overcome to become a workable, real-time assistive tool for Marathi-speaking deaf individuals. For each gap, we give an in-depth analysis of the gap and present the evidence, practical implications, and research opportunities in Table VI below.

7.1 Gap 1 (G1): Computational Overhead and Edge Hardware Incompatibility. The main obstacle to real-world adoption is the high computational cost of pixel-based deep learning architectures. The prevalent SLR models include deep 2D CNN models (VGGNet [46], ResNet50 [49], DenseNet), 3D spatio-temporal convolutions (I3D [10], [32], S3D [59]), hybrid CNN-LSTM models ([18], [20], [28]), or Vision Transformers (CvT [16], Sign Language Transformers [29] and SLRFormer [60]), as detailed in Tables I and III. The sheer number of spatial convolution parameters to evaluate is in the millions, and more than 4.5 million of those calculations is spent on parsing background textures, clothing folds and lighting gradients, not hand kinematics, when processing a 30 frame window of 224 × 224 RGB images. These models yield good accuracy at the benchmarks when they are deployed on workstation GPUs, but are severely limited by latency or thermal throttling when installed on commodity CPU-only laptops, budget smartphones and portable embedded systems [24]. On the other hand, delegating the heavy pixel models in cloud microservices [21] adds latency to the network, and does not work in low connectivity hospitals, rural schools, public offices. For a practical assistive engine, the number of input dimensions and parameters must be drastically reduced to run locally on standard CPU hardware and be able to perform inference on continuous 30 FPS.

In 7.2 Gap 2 (G2): Spatial Bias, Camera-Framing Sensitivity and Domain Shift, students will explore how camera framing affects spatial biases and how domain shift occurs. Spatial bias is a phenomenon that often occurs when models trained in controlled lab environments are used in new rooms, webcams, users etc. and has two types. First, appearance-based RGB classifiers [13]–[15], [17], [19], [46]–[48] learn the illumination profile, background colour and skin colour of the signer's face in the dataset and misclassify any data with slight variations in light sources or background noise. Second, skeletal landmark extractors like MediaPipe or OpenPose [6]–[8], [34], [50], [53], [54] are also adopted in the existing systems, which are yet to be fed with the extracted landmark raw data in the image space (x, y, z) ∈ [0, 1]3, causing severe camera-framing bias. Raw viewport coordinates indicate where the hand is within the camera frame and the distance between the signer and the lens. The numeric coordinate distribution changes dramatically even if the physical sign articulation does not change, provided that the signer steps half a meter back, leans to the side or uses a webcam that has a different field of view.

Skeletal models are fragile when applied to real-world camera locations because they lack a frame-by-frame geometric normalisation scheme with explicit translation and scale invariance.

7.3 Gap 3 (G3): Temporal Hallucinations, Absence of Idle-State Rejection. One major limitation encountered in the literature reviewed is the disconnect between the isolated test in the laboratory and the inference in the live environment. Most dynamic and video-based studies in Tables II and III [18]–[20], [51] – [55] train and evaluate their classifiers on pre-segmented video clips, with the assurance that each clip only has one valid sign. During spontaneous human interaction, however, there are frequent interruptions of the signature (idle hands in lap or at side) and of the self-adaptor (scratching, adjusting glasses, shifting body position) of the signer, hands out of the camera view frame and fast co-articulation between the next signs. If the sliding-window classifier uses a closed-set softmax output layer (with N classes), given a live webcam stream, then it is mathematically compelled to put 100% of its probability mass in those N classes on each frame. The classifier constantly hallucinates sign predictions that are confident throughout silence and ambient motion, exacerbated by the over-confidence that is common among modern deep neural networks with out-of-distribution inputs [41]. No one of the 15 primary studies is explicit in using an Idle sink class, a confidence calibration analysis and/or multi-window debounce filtering and there is no report on false activation per minute during non-signing periods.

7.4 Gap 4 (G4): Critical Regional Data Scarcity of Continuous Marathi Sign Language. Marathi Sign Language (MSL) faces severe lack of resources unlike American Sign Language (WLASL [10]) and German Sign Language (RWTH-PHOENIX-Weather 2014T [3, [4]) which have large, multi-signer video benchmarks of their own at the national level. The publicly available MSL datasets [42], [43], [46] and [50] are overwhelmingly limited to static single-frame images of the Devanagari manual alphabet and digits, recorded by a small number of participants (e.g. just four signers against a plain background in [42]). The only repository entry containing Marathi glosses at the sentence level is not peer-reviewed and is not annotated with signer-independent partitions and idle/transition class; it is not verified. Due to its regional lexical markers and the Subject–Object–Verb (SOV) syntax, training models on either ASL or static fingerspelling data is insufficient for generalisation to natural continuous Marathi sign. A documented, multi-signer continuous MSL dataset that includes stringent signer-independent splits and an explicit Idle class is thus essential for reproducible regional AT.

TABLE VI. RESEARCH GAPS, EVIDENCE AND OPPORTUNITIES

Gap	Evidence from literature	Why it matters	Research opportunity
G1. Computational overhead. Pixel-based models are too heavy for edge devices.	Hybrid CNN-LSTM, Vision Transformer and NMT approaches report heavy compute or data demands [16], [18], [20], [27]; portable systems report performance limits from scarce resources [24]; 3D CNNs process full frames [32].	Assistive technology must run on ordinary laptops and phones to reach the community it is meant for.	Replace pixel input with landmark vectors and compare model size, latency and frame rate on CPU-only hardware.
G2. Spatial bias and domain shift. Models fail on real-world hardware because of camera framing bias.	Limitations cite lighting, background clutter, occlusion and camera-angle sensitivity [13], [14], [17], [18], [19]. Public MSL data were recorded by four participants against a uniform background [42]. Raw landmark coordinates depend on position and distance from the camera.	A model that works only on the recording setup of its training data fails for a different user, camera or room.	Normalise landmarks relative to a wrist anchor, and evaluate across signers, devices and camera positions.
G3. Temporal hallucinations. Systems cannot filter ambient noise and resting states.	No primary study records idle-state rejection or a false-activation metric (Section 6.4). Closed-set softmax must always pick a class [5], and its confidence is poorly calibrated [41].	A live system that outputs signs while the signer rests or moves naturally is unusable and erodes trust.	Add an explicit Idle class, a confidence threshold and temporal debouncing, and measure false activations per minute.

Gap	Evidence from literature	Why it matters	Research opportunity
G4. Regional scarcity. There is no robust MSL dataset.	Public MSL resources are small and mostly static [42]–[43]; a sentence-level video and gloss resource is unverified [44]. Benchmarks exist for ASL [10], ISL words [9] and German SL [4].	Without a shared multi-signer, continuous benchmark, MSL results cannot be compared or reproduced.	Collect and release a multi-signer MSL corpus with documented protocols, signer-independent splits and an Idle class.

8. Future Research Directions (Proposed Architecture)

8.1 Positioning

The four gaps point to a single design target: a camera-only engine that classifies signs from landmark sequences, runs on a CPU, rejects non-signing input and is trained on MSL data. This section describes such an engine as a design hypothesis grounded in the reviewed evidence. It has not been validated in this review, and the numerical settings below are initial values to be tuned.

8.2 Zero-Pixel Extraction with Root-Anchor Normalization

The output of each webcam frame is passed to the MediaPipe Holistic [6] tracker, of which only two hand streams (with 21 3D landmarks each) are saved (Stage A of Fig. 3). No pixels are stored or transmitted, only landmark coordinates are retained and the frame is thrown away. This is the zero-pixel property: it directly targets G1 and provides a privacy gain. Each frame provides the two hands with a 126 dimensional vector (2 hands × 21 landmarks × 3 coordinates). Additional pose landmarks [8] can be added for an extension to retrieve b-body position.

To overcome G2, all landmarks are referenced to the wrist of their hand (which is the root anchor - landmark 0) and scaled to the distance from the wrist to the base of the middle finger (landmark 9) as in Eq. (1):

$$\tilde{p}_{t,h,j} = \frac{p_{t,h,j} - p_{t,h,0}}{\|p_{t,h,9} - p_{t,h,0}\|_2 + \varepsilon} \quad (1)$$

Here $p_{t,h,j} \in \mathbb{R}^3$ is the raw three-dimensional coordinate at frame $t \in \{1, ..., T\}$ for hand $h \in \{L, R\}$ and landmark index $j \in \{0, ..., 20\}$, $p_{t,h,0}$ is the wrist root anchor, $p_{t,h,9}$ is the middle-finger metacarpophalangeal (MCP) knuckle, $\|\cdot\|_2$ denotes the Euclidean norm, and $\varepsilon = 10^{-6}$ prevents division by zero. Under Eq. (1), the normalised coordinate $\tilde{p}_{t,h,j}$ is strictly invariant to in-plane translation and camera-to-signer distance scaling. When a hand is not detected in frame t, its 63 values are set to zero (zero-fill) prior to normalisation so that the 126-dimensional frame vector $\tilde{x}_t \in \mathbb{R}^{126}$ maintains a fixed shape (a zero vector remains zero under Eq. (1)). One trade-off should be evaluated empirically: anchoring each hand independently discards inter-hand distance, which can be restored by appending a 3-dimensional inter-wrist displacement vector $\Delta p_t = p_{t,R,0} - p_{t,L,0}$ or upper-body BlazePose landmarks [8].

pt,h,0 is the wrist root anchor, pt,h,9 is the middle-finger metacarpophalangeal (MCP) knuckle, ||·||2 is the Euclidean norm, and ε = 10−6 is to avoid divide by zero errors. Note that for Eq. (1), the normalised coordinate p̃t,h,j is strictly translation and camera-to-signer distance invariant to in-plane translation. A hand is not detected when the 63 values are zero, so before normalisation they are set to zero (zero-filling) and thus the frame vector x̃t, which is 126 dimensional, has a fixed shape like a zero vector under Eq. (1). The following trade-off should be measured empirically: Anchoring each hand individually would remove the inter-hand distance, which can be re-introduced by appending a 3-dimensional inter-wrist displacement vector Δpt = pt,R,0 − pt,L,0, or upper-body BlazePose landmarks [8].

8.3 Hybrid 1D-CNN + BiLSTM + Multi-Head Temporal Attention Model

The classifier is implemented on a sliding window of T = 30 normalised frames, to be input to a mathematical representation of the classifier: X ∈ ℝ30 × 126, where each frame is normalised. The three mathematical stages are selected for their specific function (Stage B of Fig. 3). Local short-range velocity and acceleration features C ∈ ℝ30 × 128 are extracted in parallel along the time axis using two stacked one-dimensional temporal conv (Conv1D) blocks [38] with K = 3, BatchNorm and Relu activation function. Second, C is processed sequentially in both the backward and forward directions by a Bidirectional LSTM [35, 36] and the directional hidden states are then concatenated to form context-enriched representations H = [h1, ..., hT] ∈ ℝ30 × 256, as defined in Eq. (2):

$$\begin{aligned} c_t &= \mathrm{ReLU}(\mathrm{BN}(W_c * \tilde{x}_t + b_c)) \\ h_t &= [LSTM_{fwd}(c_t)\,;\, LSTM_{bwd}(c_t)] \end{aligned} \quad (2)$$

Third, a multi-head temporal self-attention (MSA) module [37] with M = 4 heads (key/query dimension dk = 64 per head) calculates the scaled dot-product attention weights, with respect to the BiLSTM sequence H, such that the core manual stroke frames are given high-weight while the boundary transition and rest frames are given low-weight, followed by a residual connection, Layer Normalisation (LN), global temporal average pooling, and an (N + 1)-way softmax classifier with N+1 classes (Eq. (3)):

$$\begin{aligned} A_m &= \mathrm{softmax}\left(\frac{Q_m K_m{}^T}{\sqrt{d_k}}\right) V_m \\ \hat{y} &= \mathrm{softmax}\left(W_o \cdot \mathrm{Pool}\left(\mathrm{LN}(H + \mathrm{Concat}(A_{1,\ldots,M})W_p)\right) + b_o\right) \end{aligned} \quad (3)$$

Here Qm, Km, Vm are the linear query, key and value projections of H for attention head m, and ŷ ∈ [0, 1]N+1 is the posterior probability distribution across the N Marathi sign classes and the explicit Idle class (class index 0). The initial reference configuration is given in Table VII.

TABLE VII. INITIAL REFERENCE CONFIGURATION (ILLUSTRATIVE; TO BE TUNED)

Stage	Setting	Role
Input	30 × 126 tensor	Landmark window
1D-CNN	Conv1D, 64 and 128 filters, kernel 3, batch norm, dropout 0.3	Local temporal features
BiLSTM	128 units per direction, return sequences, dropout 0.3	Bidirectional context in the window
Temporal attention	4 heads, residual + layer norm, global average pooling	Frame weighting
Classifier	Dense layer 256 → N + 1, softmax	Sign or Idle

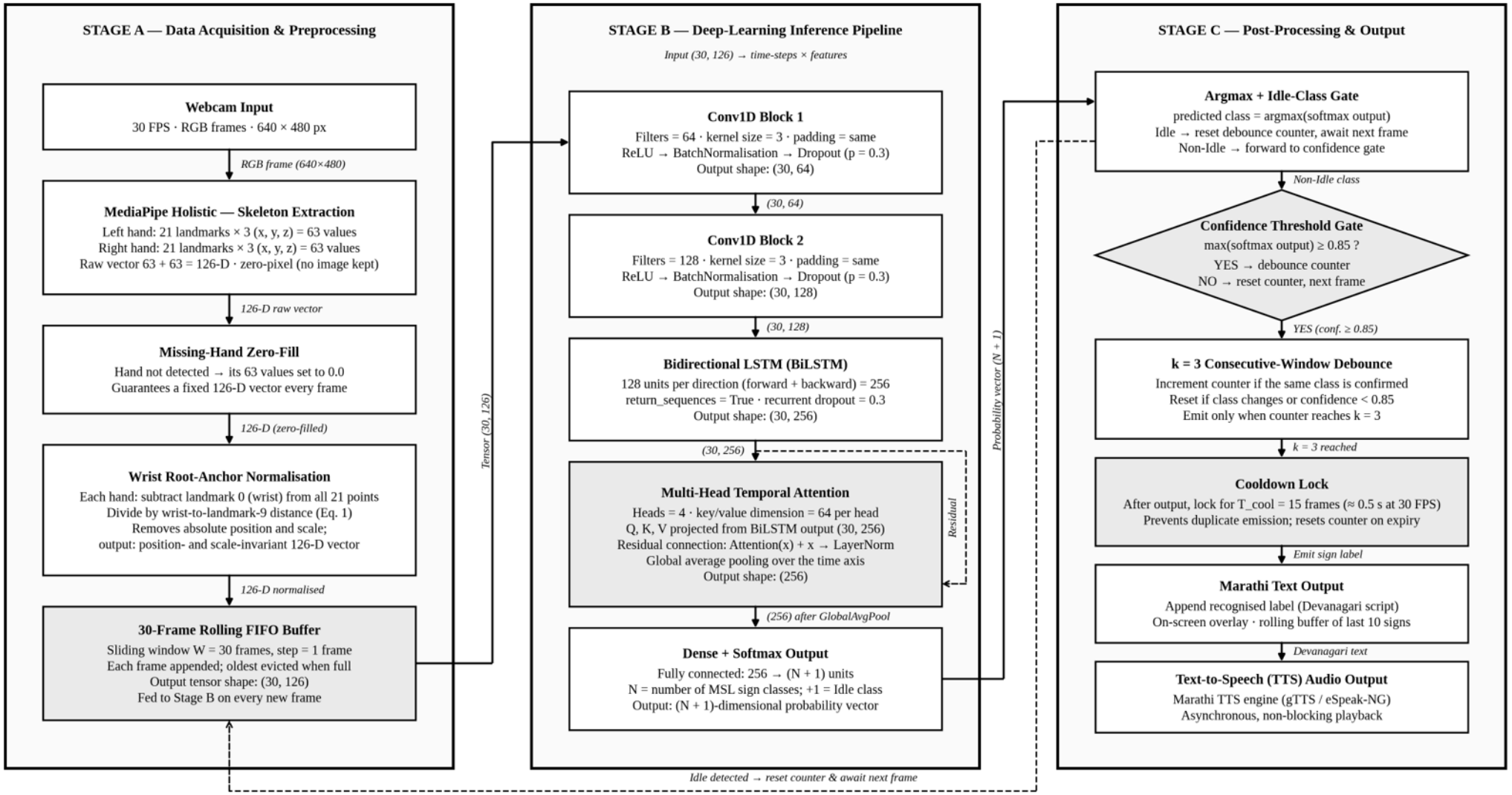


Fig. 3. Proposed camera-only system architecture: Stage A, acquisition and preprocessing; Stage B, hybrid 1D-CNN + BiLSTM + temporal-attention classifier; Stage C, post-processing and output. Dashed lines show the Idle feedback path and the attention residual connection.

8.4 Inference Control: Rolling Buffer, Idle Class and Confidence Debounce

G3 is not treated as a network, it is treated at the level of inference loop. The current landmark vectors are stored in a 30-frame (roughly 1 second) first in first out (FIFO) buffer. An explicit Idle class of sinks is trained on resting hands, hands out of frame (all-zero windows), posture adjustment, transitions and other non-sign movement, thus giving “no sign” some place to go instead of having to be forced onto a sign class. A prediction is then accepted by a confidence debounce filter if the top class isn't Idle, if the prediction probability is greater than 85%, if the same class is predicted for k windows (k = 3, for a start). Once a token is emitted, another cooldown lock (15 frames / 0.5 s at default 30 fps) will be put on that sign, which will not re-emit the token for 15 frames (the initial value), and the debounce counter will be reset when the lock finishes. When Idle is detected, the lock is also released in the live workflow in Fig. 4. This sequence is summarised in stage C of Fig. 3.

Let cw = argmaxi ŷw,i and pw = maxi ŷw,i be the predicted class and the maximum posterior probability at sliding window w, respectively. A cw sign is emitted at window w when and only when the following debounce and confidence gate (Eq. (4)) is true not in the cooldown window Tcool:

$$E_w = \mathbb{I}(c_w \neq 0) \cdot \mathbb{I}(p_w > \tau) \cdot \prod_{i=0}^{k-1} \mathbb{I}(c_{w-i} = c_w) \cdot \mathbb{I}(w \notin T_{\text{cool}}) \quad (4)$$

The indicator function I(·) is a Boolean function, with the confidence level τ = 0.85, and the number of consecutive confirming windows k = 3. The calibration of softmax confidence in deep networks can be miscalibrated [41] so τ should be calibrated on a held-out validation split [41] using a reliability diagram and Expected Calibration Error (ECE) and its sensitivity should be reported by threshold ablation (Table VIII). The whole live inference control loop, ending in Marathi Devanagari text or text to speech (TTS) synthesis is shown in Fig. 4.

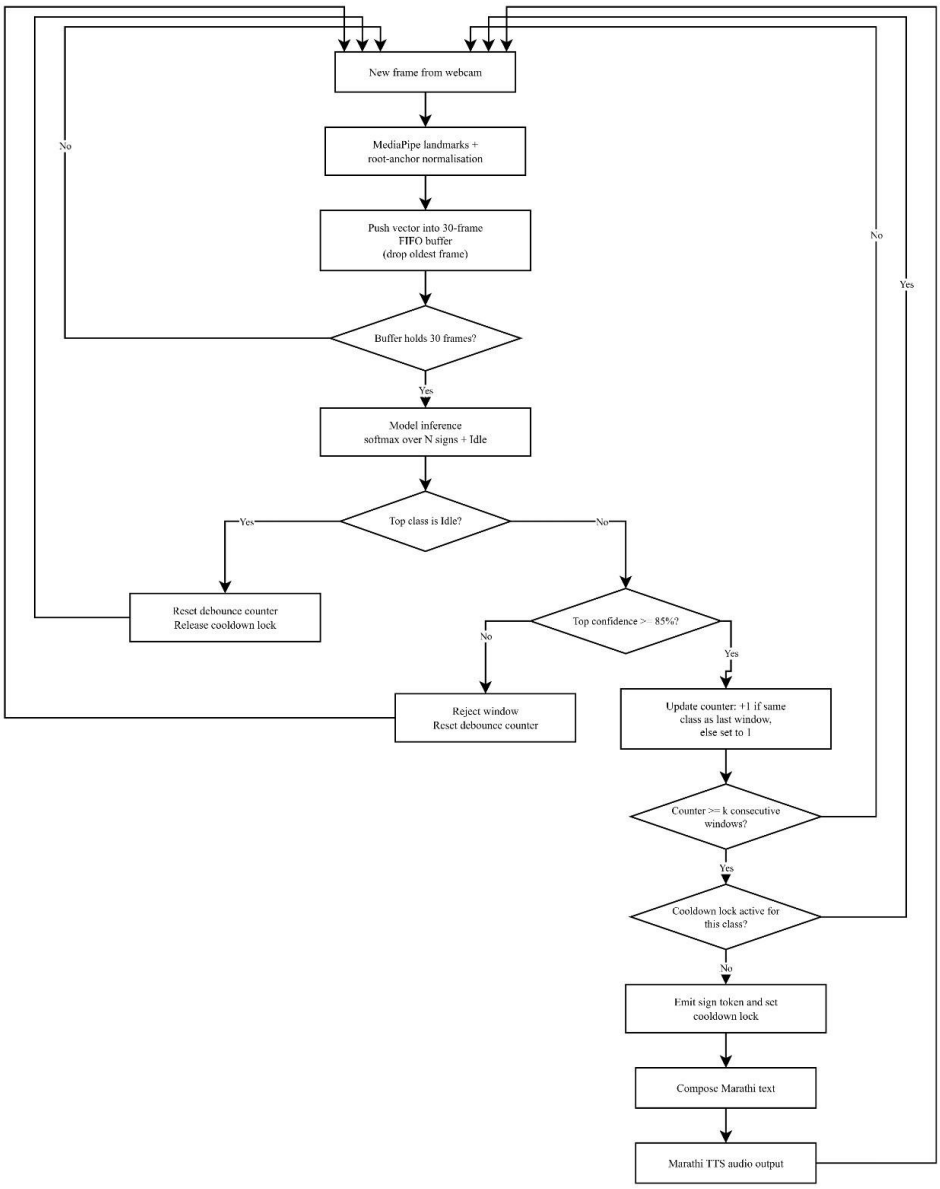


Fig. 4. Continuous live inference workflow.

8.5 Validation Agenda

The design should be evaluated against the gaps that it addresses. The experiments that are planned and the metrics used are listed below in Table VIII. Ablating each design element, which allows the contribution of the root anchor, the Idle class and the debounce filter to be measured, not guessed.

TABLE VIII. VALIDATION PLAN LINKED TO RESEARCH GAPS

Gap	Design element	Planned validation and metrics
G1	Zero-pixel landmark input; compact hybrid model	Parameter count, model size, latency per window and sustained frames per second on CPU-only hardware, against an RGB baseline.
G2	Root-anchor normalisation	Signer-independent splits; tests across webcams, distances and framings; accuracy with and without normalisation.
G3	Idle class; 85% threshold; debounce	False activations per minute on long idle and ambient-motion recordings; expected calibration error; ablation of each component; threshold sweep.
G4	New multi-signer MSL corpus	Documented collection protocol, signer-independent train/test split, per-class accuracy and macro-F1, and an explicit Idle class.

Once this baseline is validated, further directions are followed: adding face landmarks for non-manual markers, building the system to produce sign-to-speech output, exploring the system in realistic environments like clinics and classrooms with deaf users.

9. Future Scope

The proposed architecture of zero-pixel hybrid 1D-CNN + BiLSTM + Multi-Head Temporal Attention sets a lightweight baseline for privacy-preserving continuous Marathi Sign Language (MSL) recognition, while some high-impact extensions lay the groundwork for the future direction of the BhashaBridge project.

9.1 Integration of Non-Manual Markers via Holistic Facial and Upper-Body Landmarks

Manual hand shapes are often used with non-manual hand signals such as eyebrow raise or furrow, mouthing patterns, eye gaze and head tilt, as well as to differentiate between declarative statements and interrogatives, negations and conditional clauses in natural Marathi signing [2]. Future versions will append a small number of 16 MediaPipe Face Mesh keypoints (defining the eyebrows, lips, and nose tip) and 6 BlazePose upper-body landmarks (shoulders and elbows) [6] [8] to the 42 hand landmarks (126 coordinates) that are currently saved, to maximize CPU throughput. Importantly, this comprehensive landmark fusion will keep the zero-pixel privacy guarantee without compromising the possibility to disambiguate identical manual trajectories of signs with different facial grammars or with different body-anchor locations.

9.2 Gloss-to-Marathi Grammatical Translation Using On-Device Language Models

In continuous sign language, an output consists of lexical glosses without inflections (for example "मी आज रुग्णालय डॉक्टर भेटणे") while natural spoken and written Marathi is a morphologically rich, inflected language with postpositions (विभक्ती प्रत्यय), verb agreement (gender-number-person) and tense markers ("मला आज रुग्णालयात डॉक्टरांना भेटायचे आहे"). One of the key future goals of BhashaBridge is the incorporation of an offline, 4-bit quantised Small Language Model (SLM) / fine-tuned IndicBART / mT5 sequence-to-sequence module downstream of the debounce filter module to reconstruct fluent and grammatically inflected Devanagari Marathi sentences before the Text-to-Speech (TTS).

9.3 Bidirectional Communication via Speech-to-Sign 3D Avatar Synthesis

Two-way communication is essential for true accessibility in hospitals, banks and government buildings. The BhashaBridge platform will be further enhanced by implementing a reverse translation pipeline which will take the spoken Marathi from a doctor, teacher or clerk and translate it using offline Automatic Speech Recognition (ASR) into a SOV sign gloss, which will then be used to drive a lightweight, parametric 3D skeletal avatar and present continuous Marathi Sign Language (MSL) animations with smooth co-articulation transitions on screen.

9.4 Expansion to a Standardized Multi-Dialect Maharashtra MSL Benchmark Corpus

There are differences in sign vocabulary and regional lexical conventions between different regions in Maharashtra: Pune/Western Maharashtra, Mumbai/Konkan, Vidarbha, Marathwada and North Maharashtra. Further data expansion will involve working with deaf associations and special education schools throughout Maharashtra to document an open access MSL video benchmark of over 500 lexical

signs and 2,000 continuous SOV sentences by 50+ native signers, with standardized MSL partitions that are signer independent and annotated with idle and transition boundaries.

9.5 Edge Quantization, Mobile Packaging and Real-World Clinical Field Trials

Last but not least, to move from prototypes to wide-spread deployment, the trained hybrid model will be exported to ONNX Runtime and TensorFlow Lite with post-training INT8 weight quantization, with the goal of achieving sub-15 ms per-window inference on budget Android smartphones and standalone Raspberry Pi kiosk terminals. Word Error Rate (WER), False Activations Per Minute (FAPM) and System Usability Scale (SUS) scores will be assessed under real-world operation conditions using the longitudinal field evaluations with the deaf participants in the primary healthcare outpatient clinics (OPDs), municipal e-Seva kendras and inclusive classrooms at PCCoE.

10. Conclusion

To gain insights into the understanding of real time sign language recognition and gaps in the same for Marathi Sign Language, a systematic search of eight databases and supplementary skeleton based and continuous-recognition was conducted according to the guidelines in the PRISMA.A systematic search of eight databases and supplementary skeleton based and continuous recognition literature was done following the guidelines of PRISMA to understand what is known and gaps in the real time sign language recognition system for Marathi Sign Language. The main corpus consists of appearance-based CNN pipelines that are measured on private datasets in a controlled setting. Hybrid CNN-LSTM and Transformer models can be used to improve sequence modelling but require significant compute and the effect of a continuous system when the signer is not signing has not yet been reported.

There are four identified gaps: computational overhead, spatial bias and domain shift, temporal hallucination and regional data scarcity. The proposed direction is to specifically design the following to answering each question: zero-pixel landmark input, wrist-anchored normalisation, a hybrid 1D-CNN + BiLSTM + multi-head attention hybrid-classifier, a 30-frame rolling buffer with an Idle class and a calibrated confidence debounce filter, and a purpose-built MSL corpus. The validation agenda in Section 8.5 (Table VIII) and the future roadmap in Section 9 would lead to an assistive tool that can be operated by using an ordinary webcam and a personal computer, is freely usable under all lighting conditions, doesn't require any special equipment, and would enable the Marathi-speaking deaf users to have a practical communication tool in hospitals, classrooms and public offices.

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Appendix A – Search Log

The table records the per-database execution of the search strategy in Section 3.3 between 28 July and 5 August 2026. In total, 140 records were retrieved. After 18 duplicates were removed, 122 unique records were screened by title and abstract and 84 were excluded, mainly for publication before 2023, non-English language, absence of peer review, or dependence on non-camera hardware. Thirty-eight records were assessed in full text, and 23 were excluded for insufficient methodological detail, purely statistical approaches, or scope outside the four research questions, leaving the 15 studies (S1–S15) used in this review.

TABLE A1. DATABASE-WISE SEARCH LOG

Database	Date searched	Search string	Filters	Records retrieved
IEEE Xplore	28–31 Jul 2026	(“sign language recognition” OR “gesture recognition”) AND (“bidirectional translation” OR “sign-to-speech” OR “speech-to-sign”) AND (“deep learning” OR “YOLO” OR “large language model”)	2023–2026; English; Journal/Conf.	52
Google Scholar	28–31 Jul 2026	sign language recognition AND (bidirectional translation OR sign-to-speech OR speech-to-sign) AND (deep learning OR YOLO OR large language model)	2023–2026; English	30
SpringerLink	1–3 Aug 2026	(“sign language recognition”) AND (“deep learning” OR “CNN-LSTM” OR “vision transformer”) AND (“bidirectional” OR “assistive”)	2023–2026; English; Article/Review	18
ScienceDirect	1–3 Aug 2026	(“sign language”) AND (“machine learning” OR “deep learning”) AND (“translation” OR “communication system”)	2023–2026; English	14
PubMed / PMC	3–4 Aug 2026	(“sign language recognition”) AND (“assistive technology” OR “hearing impaired”) AND (“artificial intelligence”)	2023–2026; English	10
Wiley Online Library	4 Aug 2026	(“sign language recognition” OR “gesture recognition”) AND (“deep learning” OR “neural network”)	2023–2026; English	8
Nature (Scientific Reports)	4 Aug 2026	(“sign language”) AND (“recognition” OR “translation”) AND (“AI” OR “deep learning”)	2023–2026; English	5
Frontiers	5 Aug 2026	(“sign language recognition”) AND (“bidirectional” OR “assistive communication”)	2023–2026; English	3
Total				**140**